

Machine Learning | Python | SQL | EDA | Predictive Analytics | Business Intelligence

301
Vehicles
Analyzed

97.9%
RF R²
Score

 **4.66L**
Avg Resale
Price

 **10.28L**
Diesel Avg
Resale

36.6%
Avg Market
Depreciation

15 Years
Data Span
2003–2018

Model Development & Results

Model	R ² Score	MAE (■ Lakh)	Notes
Linear Regression	88.1%	~1.2L	Baseline; interpretable coefficients
Random Forest	97.9%	0.40L	Production model — saved as .pkl
Gradient Boosting	98.4%	0.30L	Highest accuracy; slower training
5-Fold CV (RF)	~97.6% avg	±0.8%	Confirms model robustness

Feature	Importance	Business Interpretation
Present_Price (ex-showroom)	86.7%	Market value anchor — strongest pricing signal
Dep_Pct (engineered)	7.35%	Value erosion rate — complements Present_Price
Car_Age (engineered)	1.81%	Newer cars retain value; validates model logic
Car_Name	1.89%	Brand premium — premium brands hold value better
Driven_KMs	1.18%	Mileage impact — less than most buyers assume

Key Business Insights

Insight	Finding	Implication
Diesel Premium	■10.28L avg vs Petrol ■3.26L (3.15x)	Stock more diesel — higher resale margin per unit
Dealer Premium	Dealer avg ■6.72L vs Individual ■0.87L	Model flags overpriced dealer listings efficiently
Auto Premium	Automatic ■9.42L vs Manual ■3.93L (2.4x)	Automatic = strong upsell signal for premium buyers
Top Value Models	Land Cruiser ■35L, Fortuner ■18.7L, Innova ■13.8L	/premium segment has highest resale floor
First Owner	0 prev owners ■4.76L vs 1 prev ■1.97L (+141%)	1st owner acquisition commands massive premium
Avg Depreciation	36.6% value loss from ex-showroom price	Buyers save avg ■2.97L vs buying new

Recommendations & Technology Stack

Audience	Recommendation	Impact
Dealers	Deploy model for instant appraisals — replace manual estimation	Eliminate mispricing from ■1.5L to ■0.40L per vehicle
Buyers	Run overpriced query: flag listings >15% above model fair value	Save ■0.5–2L per transaction
Fleet Ops	Acquire first-owner diesel vehicles in years 3–6	Optimize residual value — 3.15x petrol resale
Dev Teams	Wrap .pkl model in FastAPI endpoint	Scalable pricing at zero marginal cost per query

Category	Tools & Skills
Machine Learning	Scikit-learn — RandomForest, GradientBoosting, LinearRegression, cross_val_score
Data Analysis	Python 3.9+, Pandas, NumPy — feature engineering, EDA, correlation analysis
Visualization	Matplotlib, Seaborn — 18 charts, heatmap, actual vs predicted, feature importance
SQL	CTEs, Window Functions, CASE WHEN, RANK, DENSE_RANK, NTILE, LAG
Deployment	Pickle (.pkl), requirements.txt — production-ready model artifact